Recitation 10

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Outline

1. **Homework**

2. **Review**

3. **Recitation**

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2. Consider the same problem as in Example 4.9, but assume that the random variables X and Y are independent and exponentially distributed with different parameters λ and μ, respectively. Find the PDF of $X - Y$.

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Note: For $z < 0$, you cannot use symmetry here!

$$\begin{aligned} F_Z(z) &= \mathbf{P}(X - Y \le z) \\ &= \int_0^\infty \left(\int_{x-z}^\infty f_{X,Y}(x,y) \right) dx \\ &= \int_0^\infty \lambda e^{-\lambda x} \left(\int_{x-z}^\infty \mu e^{-\mu y} dy \right) dx \\ &= \int_0^\infty \lambda e^{-\lambda x} e^{-\mu(x-z)} dx \\ &= \frac{\lambda}{\lambda + \mu} e^{\mu z} \end{aligned}$$

4. A point (X, Y) is picked at random uniformly in the unit circle. Find the joint PDF of R and X, where $R^2 = X^2 + Y^2$.

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Note 1: R is not uniformly distributed.

$$F_R(r) = \frac{\pi r^2}{\pi} = r^2$$
$$f_R(r) = \frac{d}{dr} F_R(r) = 2r$$

Note 2: If you use Jacobian, be aware that $(R, X) = (r, x)$ maps to two points $(X, Y) = (x, \pm\sqrt{r^2 - x^2})$. Therefore,

$$f_{R,X}(r, x) = f_{X,Y}(x, y) \cdot 2 \left| \det\left(\frac{\partial(x, y)}{\partial(r, x)}\right) \right| = \frac{2r}{\pi\sqrt{r^2 - x^2}}$$

Note 3: If you first calculate CDF and then differentiate to get PDF, the integration in CDF is not necessary to be computed.

$$
\begin{aligned}
F_{R,X}(r,x) &= \mathbf{P}(R \le r, X \le x) \\
&= \int_{-r}^{x} \left(\int_{-\sqrt{r^2-u^2}}^{\sqrt{r^2-u^2}} f_{X,Y}(u,v)dv \right) du \\
&= \int_{-r}^{x} \frac{1}{\pi} \cdot 2\sqrt{r^2-u^2} du
\end{aligned}
$$

$$
\begin{aligned}
f_{R,X}(r,x) &= \frac{\partial^2}{\partial r \partial x} F_{R,X}(r,x) \\
&= \frac{\partial}{\partial r} \left(\frac{1}{\pi} \cdot 2\sqrt{r^2-x^2} \right) \\
&= \frac{2r}{\pi\sqrt{r^2-x^2}}
\end{aligned}
$$

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- Conditional expected value

$$\mathbf{E}[X|Y=y]=\sum_x x p_{X|Y}(x|y)$$

$$\mathbf{E}[\mathbf{E}[X|Y]]=\sum_y \mathbf{E}[X|Y=y]p_Y(y)=\mathbf{E}[X]$$

- Conditional variance

$$\text{var}(X \mid Y)=\mathbf{E}\big[(X-\mathbf{E}[X \mid Y])^2 \mid Y\big]=\mathbf{E}[\tilde{X}^2 \mid Y]$$

$$\text{var}(X)=\mathbf{E}[\text{var}(X \mid Y)]+\text{var}(\mathbf{E}[X \mid Y])$$

- Transforms

$$M_X(s)=\mathbf{E}[\mathrm{e}^{sX}]$$

$$\mathbf{E}[X^n]=\left.\frac{\partial^n M_X(s)}{\partial s^n}\right|_{s=0}$$

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1. Let X, Y and Z be discrete RVs. Show the following generalizations of the law of iterated expectation.

(a) $\mathbf{E}[Z] = \mathbf{E}\big[\mathbf{E}[Z \mid X, Y]\big]$.

(b) $\mathbf{E}[Z \mid X] = \mathbf{E}\big[\mathbf{E}[Z \mid X, Y] \mid X\big]$.

(c) $\mathbf{E}[Z] = \mathbf{E}\Big[\mathbf{E}\big[\mathbf{E}[Z \mid X, Y] \mid X\big]\Big]$.

Proof:

(a)

$$
\begin{aligned}
\mathbf{E}\big[\mathbf{E}[Z \mid X, Y]\big] &= \sum_x \sum_y \mathbf{E}[Z \mid X = x, Y = y] p_{X,Y}(x, y) \\
&= \sum_x \sum_y \sum_z z\, p_{Z|X,Y}(z|x, y) p_{X,Y}(x, y) \\
&= \sum_x \sum_y \sum_z z\, p_{X,Y,Z}(x, y, z) \\
&= \sum_z z\, p_Z(z) \\
&= \mathbf{E}[Z]
\end{aligned}
$$

(b)

$$
\begin{aligned}
\mathbf{E}[\mathbf{E}[Z \mid X, Y] \mid X = x] &= \sum_y \mathbf{E}[Z \mid X = x, Y = y] p_{Y|X}(y|x) \\
&= \sum_y \sum_z z p_{Z|X,Y}(z|x, y) p_{Y|X}(y|x) \\
&= \sum_y \sum_z z p_{Z,Y|X}(z, y|x) \\
&= \sum_z z p_{Z|X}(z|x) \\
&= \mathbf{E}[Z \mid X = x]
\end{aligned}
$$

Since this is true for all possible values of x, we have $\mathbf{E}[Z \mid X] = \mathbf{E}[\mathbf{E}[Z \mid X, Y] \mid X]$.

(c) Taking expectations for both sides of the formula in part (b), we obtain

$$\mathbf{E}\Big[\mathbf{E}[\mathbf{E}[Z \mid X, Y] \mid X]\Big] = \mathbf{E}[\mathbf{E}[Z \mid X]] = \mathbf{E}[Z].$$

2. Widgets are stored in boxes, and then all boxes are assembled in a crate. Let X be the number of widgets in any particular box, and N be the number of boxes in a crate. Assume that X and N are independent integer-valued RVs, with expected value equal to 10, and variance equal to 16. Evaluate the expected value and variance of T , where T is the total number of widgets in a crate.

Solution:

Let X_i denote the number of widgets in the i-th box, then $T = \sum_{i=1}^{N} X_i$. We have

$$\begin{aligned}
\mathbf{E}[T] &= \mathbf{E}\Big[\mathbf{E}\big[\sum_{i=1}^{N} X_i \mid N\big]\Big] \\
&= \mathbf{E}\Big[\sum_{i=1}^{N} \mathbf{E}[X_i \mid N]\Big] \\
&= \mathbf{E}\Big[\sum_{i=1}^{N} \mathbf{E}[X]\Big] \\
&= \mathbf{E}[N]\mathbf{E}[X].
\end{aligned}$$

Since

$$\mathbf{E}[T|N] = N\mathbf{E}[X]$$
$$\text{var}(T|N) = N\text{var}(X)$$

By the law of total variance,

$$\begin{aligned}\text{var}[T] &= \mathbf{E}[\text{var}(T|N)] + \text{var}(\mathbf{E}[T|N]) \\ &= \mathbf{E}[N]\text{var}(X) + (\mathbf{E}[X])^2\text{var}(N) \\ &= 160 + 100 \times 16 \\ &= 1760\end{aligned}$$

3. A building has N floors above the ground floor. The number of people entering an elevator on the ground floor is a Poisson random variable with parameter λ. Each person is equally likely to get off at any one of the N floors, independently of where the others get off. Please compute the mean and variance of stops that the elevator will make before discharging all of its passengers.
Hint: Using Taylor expansion $e^x = \sum_{k=0}^{\infty} x^k/k!$ to simplify your result.